SYSTEMS MANAGEMENT



System-Level Manageability for GPU Platforms

Progress in the System GPU Management Workstream

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Agenda

- From hyperscale contributions to an open system-level workstream
- System GPU Management Workstream
  - OCP GPU specifications review and feedback
  - Prescriptive profile for managing the OCP UBB
  - Redfish Message registries for emitted messages
  - Aggregation Modeling in Redfish
  - FRU Format

UBB = Universal Base Board

From hyperscale contributions to an open system-level workstream

Hyperscale GPU Mgmt WS (2023)

- Charter
  - Management of the hyperscale GPUs
- Semi-private
  - Meetings and documents under CLA
  - Contributions to OCP are public
- Wiki (contributed documents)
- Workstreams
  - FW Update
  - Manageability Interfaces & Telemetry
  - RAS

System GPU Mgmt WS (2025)

- Charter
  - System level management of GPUs
- Public
  - Co-leads: Hari Ramachandran, John Leung
  - Contributions are Common Creative licensing
- Wiki (agenda/minutes, recordings, presentations)
- GitHub repository (draft docs, profile)

OCP GPU Specifications Review

1 REVIEW & COMMENT

- Reviewed the contributed OCP GPU Management specifications in the public System GPU Management workstream.
- Consolidated implementation-focused comments on clarity, consistency, and interoperability.

2 NORMATIVE LENS

- Created a normative draft and traceability view to make requirements explicit and expose ambiguity.
- Used the exercise as a review tool—not as a competing specification.

3 ONE-SPEC PATH

- Aligned on evolving one specification rather than maintaining parallel normative and descriptive versions.
- Wait for the forthcoming revision, then verify how the review feedback improved clarity.

IMPACT: Normative review feedback is shaping a clearer, more implementable next revision.

OCP UBB interoperability profile evolution

- V0.9.1 - contribution
  - Profile contributed by the Hyperscale GPU Mgmt workstream (2025)
  - Established a public baseline for GPU-based UBB manageability
- V1.x – OCP governance
  - Profile published in the HW Management profile repository
  - System GPU Mgmt workstream assumes governance and handles issues (2026)
- V2.0 – Redfish alignment
  - Adopted MultipartHttpPushUri for firmware update (PR #60)
  - Updated ownership and contact properties

Redfish Message Registries

1 DEFINE THE GAP

- Agreed that GPU management needs a prescribed subset of existing Redfish messages plus new messages where gaps remain.
- Focused on accelerator, scale-up fabric, health, and operational events.

2 CONVERGE THE MODEL

- Use AcceleratorDevice semantics instead of NetworkDevice.
- Extend the Fabrics registry where practical; create accelerator-specific messages when needed.
- Organize messages as Critical, Warning, and OK (PR #55).

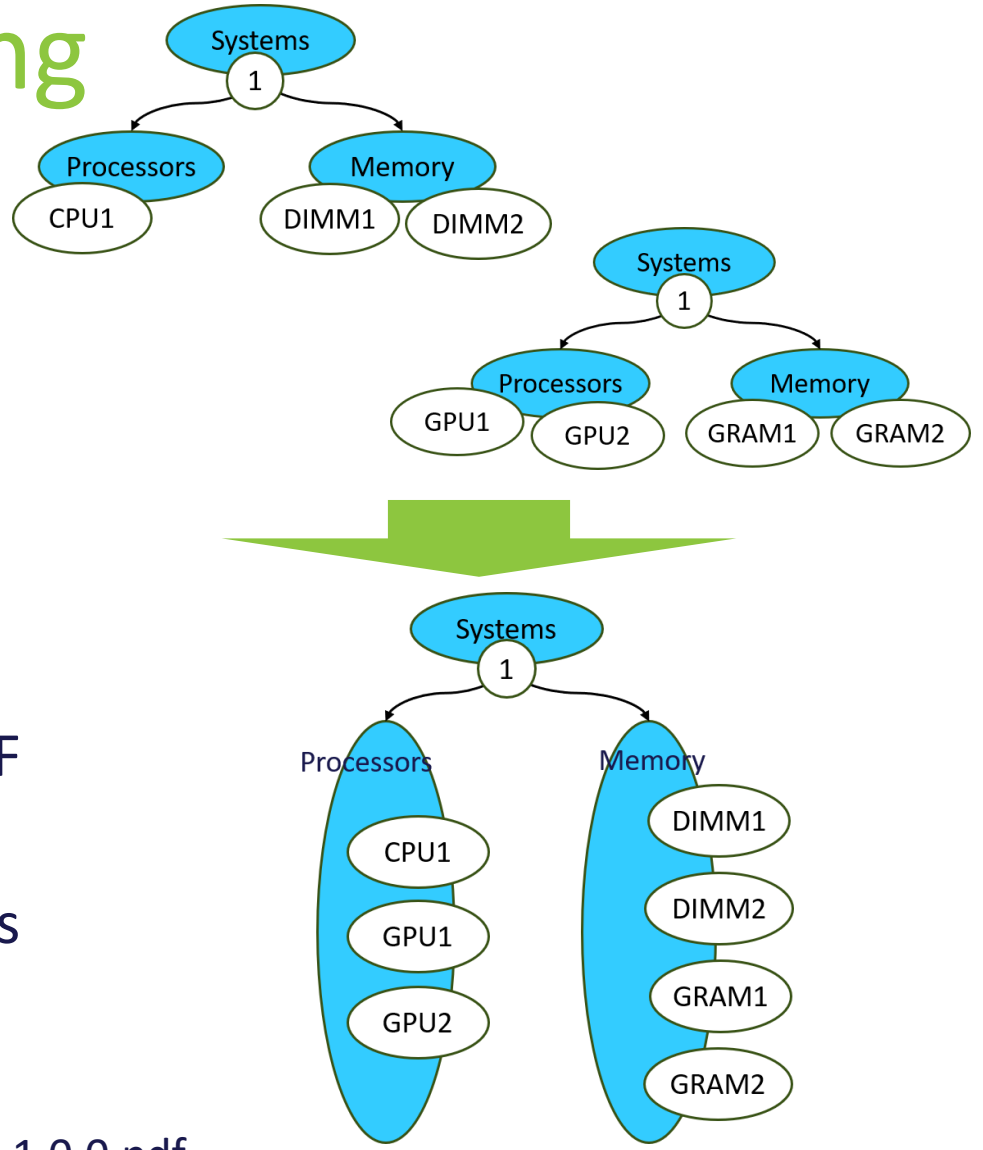
3 STANDARDS PATH

- Workstream refined the problem statement and reviewed proposals
- Engaged DMTF and incorporated feedback.
- Converting the Fabric registry to JSON for review (Issue #56)

IMPACT: Shared GPU event semantics with a concrete path from OCP collaboration to Redfish standardization.

Consistent Aggregation Modeling

- Problem
  - Redfish model allows implementors to represent systems containing GPUs and CPUs in different ways
  - Different resources hierarchies force client-specific logic and reduce interoperability
- Solution
  - Apply consistent GPU modeling guidance from DMTF DSP2090
  - Cover GPU systems and compute expansion modules



https://www.dmtf.org/sites/default/files/standards/documents/DSP2090_1.0.0.pdf

FRU Data: Preparing for the 2027 Rollover

1 IMMINENT PROBLEM

- The IPMI FRU manufacturing-date field will wrap around in November 2027
- OpenBMC has a temporary decode workaround

2 COMPATIBILITY BRIDGE

DMTF DSP0220 allows modern and legacy IPMI FRU records to coexist

3 DRAFT DIRECTION

- Propose coexistence of IPMI and DMTF FRU formats (PR #58)
- Migrate access from legacy IPMB to PLDM Type 4 (DMTF DSP0257)

IMPACT: A backward-compatible path away from a known IPMI FRU date limit before it affects new GPU platforms.

Call to Action

- Bring system-level issues to the System GPU Management workstream
- Use the UBB v2.0 Profile your implementations and provide feedback
- Where to find additional information
  - Wiki (meeting agenda/minutes/recordings, presentations)
  - GitHub repository (draft docs, profile)

Wiki – https://github.com/opencomputeproject/ocp-hm-system-gpu-management/wiki
GitHub - https://github.com/opencomputeproject/ocp-hm-system-gpu-management

Thank You!

